



MANIPAL ACADEMY OF HIGHER EDUCATION

ALGEBRAIC STRUCTURES [MAT 2235]

Marks: 50

Duration: 180 mins.

Descriptive Type Questions

Answer all the questions.

Section Duration: 180 mins

- 1) Let G be a group. Solve simultaneously, $x^2a = bxc^{-1}$ and $acx = xac$ for all $x, a, b, c \in G$. (3)
 - A)
 - B) Let $U(n)$ be the set of positive integers less than n and relatively prime to n ($n > 1$). Find the order of each element in the multiplicative group $U(15)$. (3)
 - C) Let $a \neq 0$ be an element in a ring. Prove that a cannot be both a unit and a zero-divisor. Find all zero-divisors in the ring $(\mathbb{Z}_{36}, +, \cdot)$. (4)
- 2) Prove that a field is an integral domain. Is the converse true? Justify. Explain whether $\mathbb{Q}[\sqrt{2}]$ is a field. (3)
 - A)
 - B) Find all generators and all subgroups of additive group $\mathbb{Z}_{48}$. (3)
 - C) Explain in detail why $\mathbb{Z}_3[x]/\langle x^2 \rangle$ is not a field. (4)
- 3) Let $f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x, y) = x + y$. Verify whether f is a homomorphism. If so, find its kernel. (3)
 - A)
 - B) Let G be a group acting on a set X . Then prove that the set of all orbits in X under G is a partition of X , and for any $x \in X$, the map $\varphi: Gx \rightarrow G/G_x$ is a bijection. (3)
 - C) Prove that the centre of a group G is a subgroup of G . Is it normal? What is the centre of the symmetric group S_3 on three symbols? (4)
- 4) Find all solutions of $x^2 - 7x + 12 = 0$ in the rings $(\mathbb{Z}_{13}, +, \cdot)$ and $(\mathbb{Z}_{12}, +, \cdot)$. (3)

A)

B) Determine if each polynomial is factorable or un-factorable in the specified rings.

$$1. x^{273} + 3x^{152} + 5x^{17} + 10 \text{ in } \mathbb{Z}_{19}[x]. \quad (3)$$

$$2. x^3 + x + 1 \text{ in } \mathbb{Z}_5[x].$$

C) Let R be an integral domain and $f(x), g(x) \in R[x]$ with $f(x) \neq 0 \neq g(x)$. Then prove that $\deg(f(x) \cdot g(x)) = \deg f(x) + \deg g(x)$. Give a **non-example** of a field. (4)

5) Express each of the following permutations in S_9 as a product of transpositions.

A) 1.
$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 7 & 9 & 5 & 3 & 1 & 2 & 4 & 8 & 6 \end{pmatrix} \quad (3)$$

2.
$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 9 & 8 & 7 & 4 & 3 & 6 & 5 & 1 & 2 \end{pmatrix}$$

B) Define (i) Extension Field (ii) Algebraic Element and give two examples of each. (3)

C) Let G be a commutative group $\{e, a, b, c, ab, bc, acc, abc\}$ whose generators satisfy: $a^2 = b^2 = c^2 = e$. Write the table for G . (4)

-----End-----

2a. ~~Q1~~ To prove, R be an integral domain

Let $\alpha, \beta \in R$ such that $\alpha\beta = 0$

To show, $\alpha = 0$ or $\beta = 0$

Assume, $\alpha \neq 0$

$$\text{Now, } \alpha \cdot \beta = \alpha \cdot 0$$

$$\Rightarrow \alpha^{-1} \alpha \beta = \alpha^{-1} \alpha \cdot 0$$

$$\Rightarrow \beta = 0, \text{ since } \alpha \neq 0, \alpha^{-1} \text{ exists, } \alpha^{-1} \text{ is a unit}$$

$\therefore R$ is an integral domain.

Converse is not true. $\mathbb{Z}$ is an ID not a field.

$$\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$$

Let $a, b \in \mathbb{Q}$ such that $a + b\sqrt{2} \in \mathbb{Q}[\sqrt{2}]$, then

$$1 = (a + b\sqrt{2}) \frac{(a - b\sqrt{2})}{(a + b\sqrt{2})(a - b\sqrt{2})}$$

$$= (a + b\sqrt{2}) \frac{(a - b\sqrt{2})}{a^2 - 2b^2}, \text{ and since } a^2 - 2b^2 \in \mathbb{Q} \text{ and}$$

$a - b\sqrt{2} \in \mathbb{Q}(\sqrt{2})$, this implies that $\frac{a - b\sqrt{2}}{(a^2 - 2b^2)}$ is inverse of $a + b\sqrt{2}$

$\Rightarrow \mathbb{Q}(\sqrt{2})$ is ~~not~~ a field.

2b. Generators of $\mathbb{Z}_{48} \rightarrow 1, 5, 7, 11, 13, 17, 19, 23, 25, 29, 31, 35, 37, 41, 45, 47$

Subgroups of $\mathbb{Z}_{48} \rightarrow \langle 1 \rangle, \langle 2 \rangle, \langle 3 \rangle, \langle 4 \rangle, \langle 6 \rangle, \langle 8 \rangle, \langle 12 \rangle, \langle 16 \rangle, \langle 24 \rangle, \langle 48 \rangle$

$$2c. \mathbb{Z}_3[x] / \langle x^2 \rangle = \{(ax + b) + \langle x^2 \rangle \mid a, b \in \mathbb{Z}_3\}$$

$$= \{0 + \langle x^2 \rangle, 1 + \langle x^2 \rangle, 2 + \langle x^2 \rangle, x + \langle x^2 \rangle, (x+1) + \langle x^2 \rangle, (x+2) + \langle x^2 \rangle, 2x + \langle x^2 \rangle, (2x+1) + \langle x^2 \rangle, (2x+2) + \langle x^2 \rangle\}$$

$$\textcircled{1} \text{ If possible, } (2x+1) + \langle x^2 \rangle = (x+2) + \langle x^2 \rangle$$

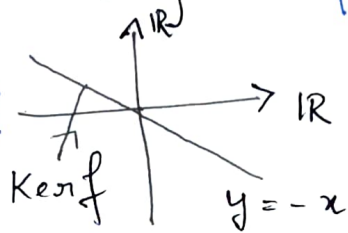
$$\Rightarrow (2x+1) - (x+2) = (x-1) \in \langle x^2 \rangle, \text{ a contradiction}$$

$$\textcircled{2} ((x+1) + \langle x^2 \rangle) \cdot ((2x+1) + \langle x^2 \rangle) = 1 + \langle x^2 \rangle$$

$$\textcircled{3} \text{ Now, } (x + \langle x^2 \rangle) \cdot (x + \langle x^2 \rangle) = x^2 + \langle x^2 \rangle = 0 + \langle x^2 \rangle$$

Therefore $x + \langle x^2 \rangle$ is not a unit, it is a zero-divisor. $\therefore \mathbb{Z}_3[x] / \langle x^2 \rangle$ not a field

$f(x, y) = x + y$ is a homomorphism.

$$\begin{aligned} \text{Ker } f &= \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid f(x, y) = 0_{\mathbb{R}}\} \quad \text{--- 1M} \\ &= \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid x + y = 0_{\mathbb{R}}\} \quad \left. \begin{array}{l} \text{1M} \\ \text{1M} \end{array} \right\} \\ &= \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y = -x\} \\ &= \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y = -x\} \end{aligned}$$


3b. For any $x, y \in X$.

Let $x \sim y \Rightarrow x = ay$ for some $a \in G$.

$\sim$ is an equivalence relation on X , and the equivalence class of $x \in X$ is the orbit Gx . Hence, the set of all orbits is a partition of X . Therefore,

$X = \bigcup_{x \in C} Gx$ (disjoint), $C \subseteq X$ containing exactly one element from each orbit.

Given, $x \in X$, consider the mapping

$\phi: Gx \rightarrow G/Gx$, given by $ax \mapsto aGx, \forall a \in G$

prove ϕ is well-defined and injective. Moreover ϕ is obviously surjective, so ϕ is bijective.

3c. $e \in Z(G)$

Take $a, b \in Z(G)$

Consider, $(ab)x = a(bx) = a(xb) = (ax)b = (xa)b = x(ab)$

$\Rightarrow ab \in Z(G)$

Consider, $ax = xa$

$\Rightarrow a^{-1}(ax)a^{-1} = a^{-1}(xa)a^{-1}$

$\Rightarrow xa^{-1} = a^{-1}x$

$\Rightarrow a^{-1} \in Z(G)$

$\therefore Z(G) \leq G$

Centre of a group G is always normal.

$Z(S_3) = \{e\}$

4a. $\mathbb{Z}_{13}$ is a field

So the possible solutions of $x^2 - 7x + 12 = 0$
is $x = 3, 4$. 1 1/2 M

In $\mathbb{Z}_{12}$, $x^2 - 7x + 12 = 0$ reduces to $x(x-7) = 0$.

Possible solutions $\rightarrow 0, 7, 3, 4$ 1 1/2 M

4b. ① When $x = 1$, $f(1) = 1 + 3 + 5 + 10 \pmod{19}$
 $= 0$.

$\therefore$ Factorable, since root exists, with explanation 1 1/2 M

② $f(0) = f(1) = f(2) = f(3) = f(4) \neq 0$

$\therefore$ Un factorable, since root doesn't exist.
with explanation 1 1/2 M

4c. If R has no zero divisors then $R[x]$ also has
no zero-divisors, 1 M

If $p(x)$ and $q(x)$ are polynomials with leading
coefficients $a_n x^n$ and $b_m x^m$, respectively, then the
leading term of $p(x)q(x)$ is $a_n b_m x^{n+m}$ & $a_n b_m \neq 0$. 1 M

$\mathbb{Z}$ is a non-example of a field, it is an
integral domain. 2 M

5a.

$$\textcircled{1} (17435)(296)(8)$$

$$= (15)(13)(14)(17)(26)(29)$$

} — 1½ M

$$\textcircled{2} (1928)(375)(4)(6)$$

$$= (18)(12)(19)(35)(37)$$

} — 1½ M.

5b.

i) If K is a field containing the subfield F , then K is said to be an extension field (or simply an extension) of F , denoted by K/F or by the diagram



} 1½ M

Ex:- $\mathbb{C}$ is an extension field of $\mathbb{R}$

ii) Let F be a field and let K be an extension of F . The element $\alpha \in K$ is said to be algebraic over F if α is a root of some non-zero polynomial $f(x) \in F[x]$.

} 1½ M.

Any other example is OK.

Example $\rightarrow \sqrt{2}$ is algebraic over $\mathbb{Q}$.

5c.

	e	a	b	c	ab	bc	ac	abc
e	e	a	b	c	ab	bc	ac	abc
a	a	e	ab	ac	b	abc	c	bc
b	b	ba	e	bc	a	c	abc	ac
c	c	ac	bc	e	abc	b	a	ab
ab	ab	b	a	abc	e	ac	bc	c
bc	bc	abc	c	b	ac	e	ab	a
ac	ac	c	abc	a	bc	ab	e	b
abc	abc	bc	ac	ab	c	a	b	e

4 M.

Row/col wise marking.